

Faithful, Interpretable Chest X-ray Diagnosis with Artifact-free B-cos Networks

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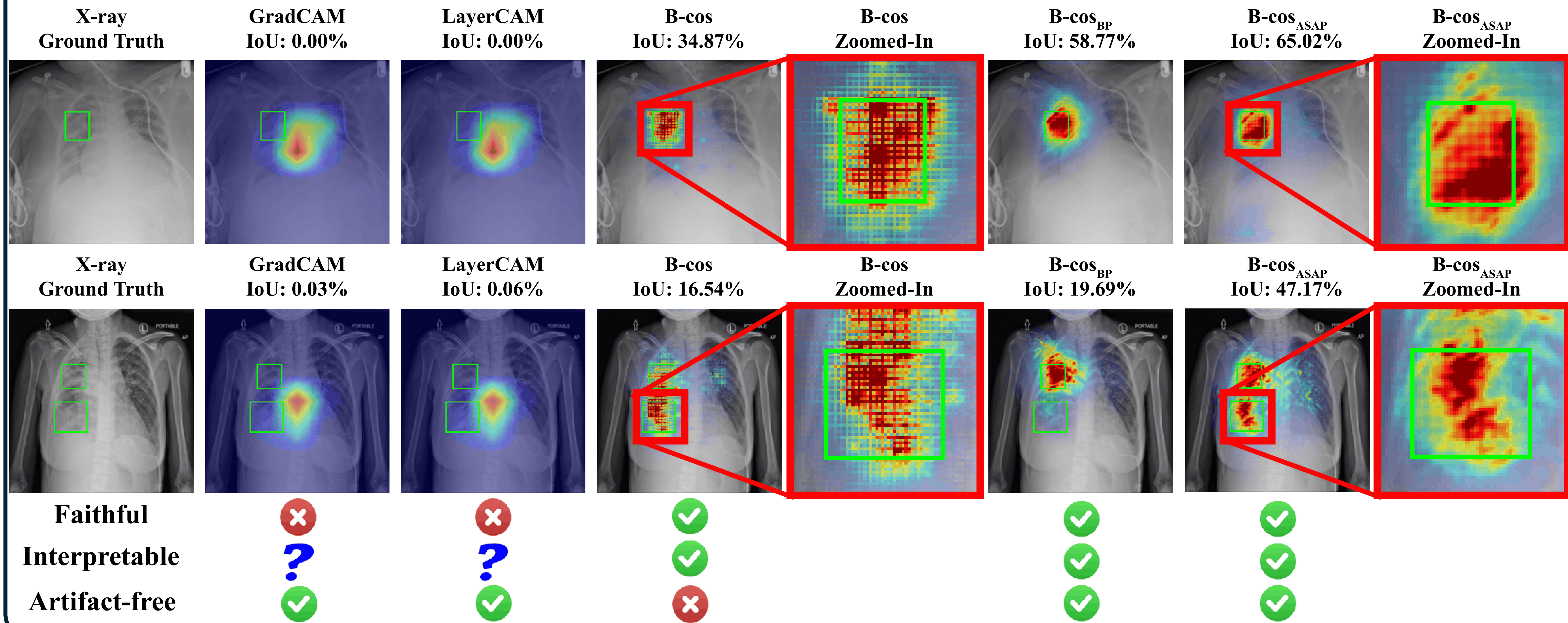
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MOTIVATION



METHOD

B-cos models

$$B\text{-cos}(x; w) = \hat{w}^\top x |c(x, \hat{w})|^{B-1},$$

$$\hat{w} = \frac{w}{\|w\|}, \quad c(x, \hat{w}) = \frac{\hat{w}^\top x}{\|x\|}$$

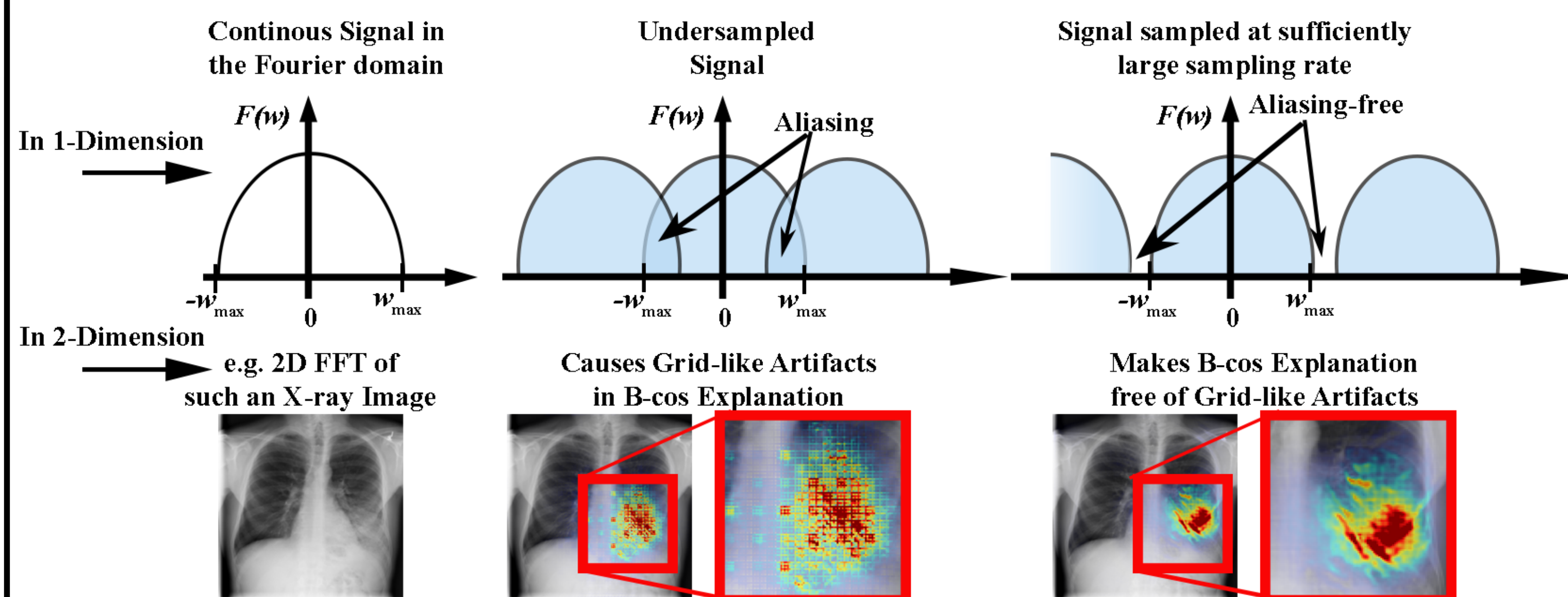
How to get the explanations?

$$s_j^l(x) = [W_{1 \rightarrow l}(x)]_j^\top \odot x$$

$$s_j^l(x) = \frac{\partial f_j(x)}{\partial x} \odot x = [W_{1 \rightarrow l}(x)]_j^\top \odot x$$

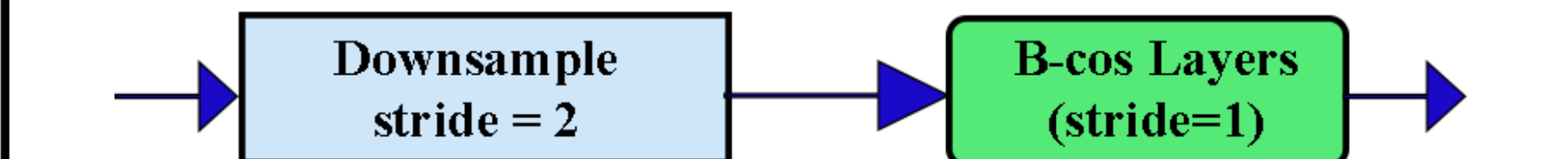
explanation mode

Motivation

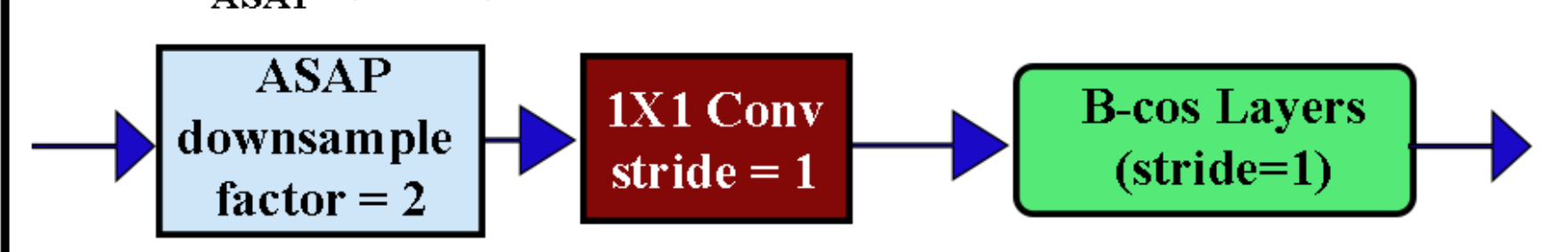


Implementation

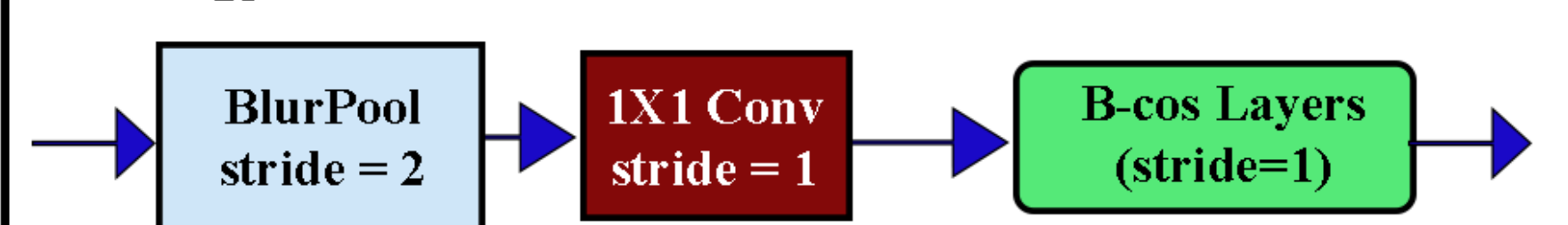
Vanilla B-cos:



B-cos_{ASAP} (ours):

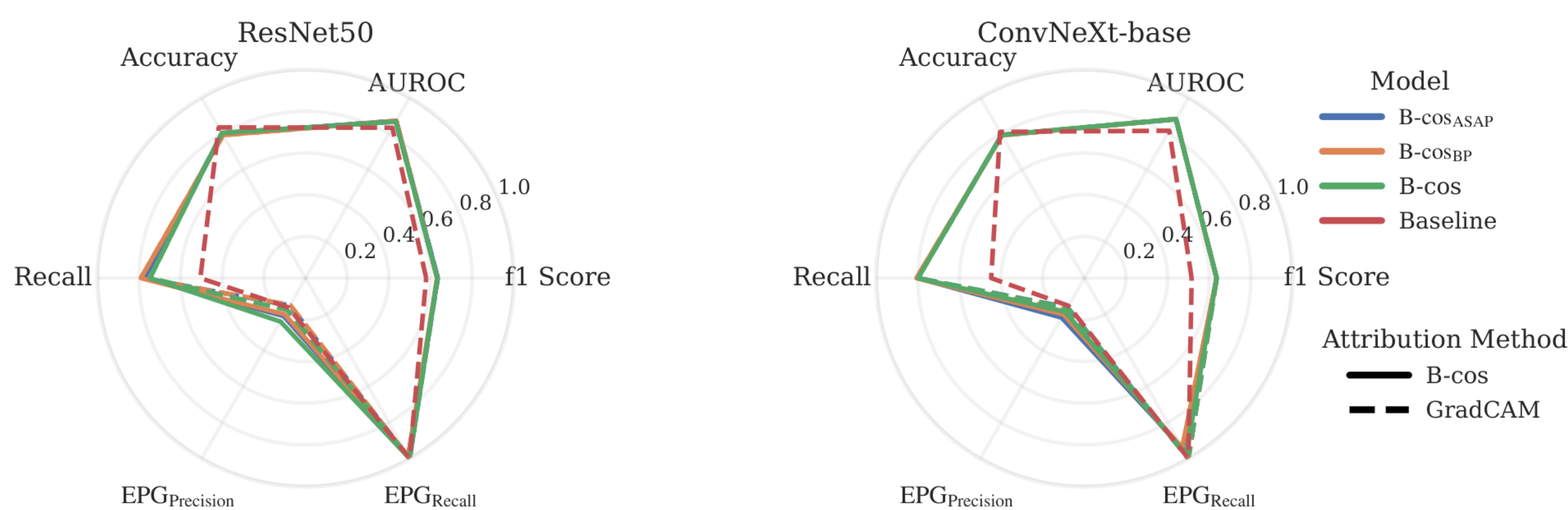


B-cos_{BP} (ours):

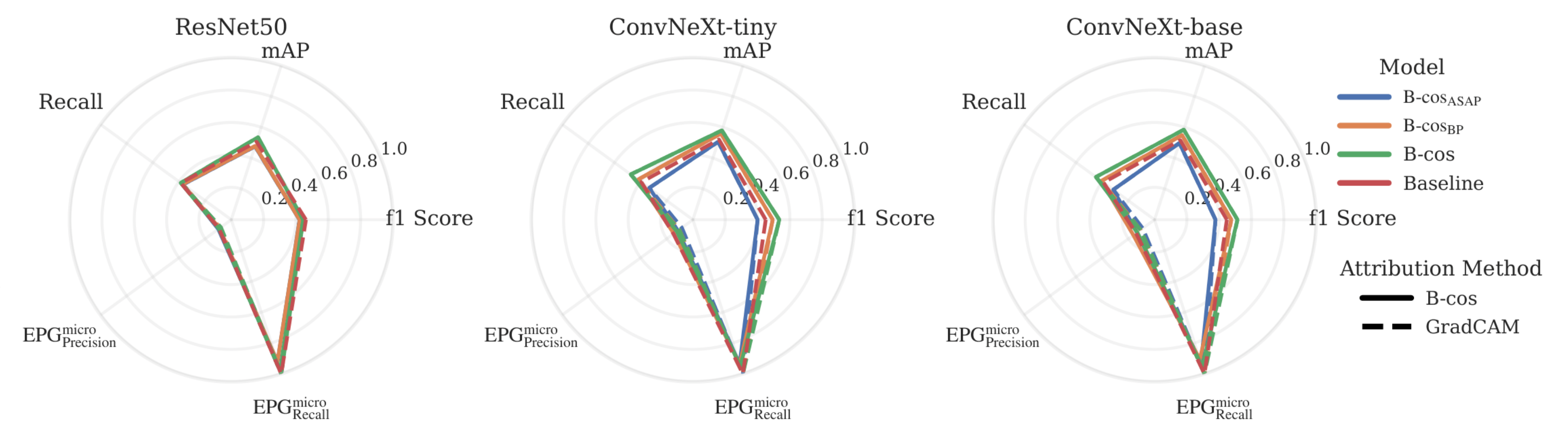


RESULTS

RSNA Pneumonia Detection Challenge



VinBigData Chest X-ray Abnormalities Detection dataset



CONCLUSIONS

- Faithful by design:** B-cos networks provide built-in, class-specific explanations for multi-class and multi-label chest X-ray diagnosis, without post-hoc attribution.
- Artifact-free explanations:** ASAP and BlurPool remove downsampling-induced grid artifacts, improving visual clarity and localization of disease-relevant regions.
- Strong diagnostic performance:** Competitive results on RSNA and VinBigData, with approximately **80% pneumonia recall versus 45%** for baseline ConvNeXt-base.

REFERENCES

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